

Block Explorer: The Scratch Toolbox

Every colour has a job. Explore the blocks that make a Scratch project move, react, remember, and shine.

We will meet the whole team.

Motion · Looks · Sound · Events · Control · Sensing ·
Operators · Variables · Lists · My Blocks · Extensions



Every colour has a superpower.

You do not need to memorise every block today. First, learn which team to visit when you have an idea.

MOVE & SHOW

Motion • Looks • Sound

START & RUN

Events • Control

NOTICE & THINK

Sensing • Operators

REMEMBER & INVENT

Variables • Lists • My Blocks • Extensions



Make the audience see the action.

Motion blocks move a sprite. Looks blocks help the sprite react. Together they turn an idea into a scene.

when green flag clicked

move 10 steps

switch costume to “happy”

Try it: change the number of steps. What happens when you use 2, 20, or -10?



Start a moment. Then make it pop.

GREEN FLAG
starts the project

KEY PRESSED
starts an action

SPRITE CLICKED
starts a surprise

A TINY MUSIC MOMENT

when this sprite clicked

start sound “pop”

say “Nice click!” for 2 seconds

Events tell Scratch when to begin. **Sound** lets your project celebrate, warn, or sing.



Be the director of your code.

Control blocks tell a script what to do next: repeat, pause, or choose a path.



repeat

Do the same action again and again.



wait

Leave space before the next action.



if / else

Choose what happens when a condition is true or false.

A REPEATING DANCE

when green flag clicked

repeat 4

turn clockwise 90 degrees

wait 0.3 seconds





Sensing is noticing. Your game can listen for a key, check a touch, watch the mouse, or remember an answer.

06 / 12 • SENSING

Let your project pay attention.

A game feels interactive when it can notice what the player is doing.

Keyboard

Is the space key pressed?

Touching

Is the sprite touching the star?

Mouse

Where is the mouse pointer?

Question

What answer did the player type?

```
if < touching [star] ? > then  
  play sound [cheer]  
  change [score] by 1
```

Give your project a thinking brain.

Green Operator blocks work with numbers, words, and true-or-false questions.



Do maths

`5 + 3`

Compare

`score > 10`

Join words

`join "Hi" name`

Make it random

`pick random 1 to 10`

Game maker tip: A random block keeps a game surprising. Try using it to choose where a star appears next.

Give your project a memory.

Variable: one changing value.
List: a collection in order.

Variables keep one fact.

Use a variable when you want to remember something that changes, like a score, timer, or number of lives.

score = 7

When the player catches a star: change score by 1.

Lists keep many facts.

Use a list when you want to store several things, like favourite colours, quiz questions, or collected items.

apple

mango

banana

Add or remove an item as the story grows.

Quick choice: Is there one score that changes? Choose a variable. Are there many prizes to remember? Choose a list.



Invent a helper for repeated actions.

A **My Block** is your own reusable instruction. Define it once. Use it whenever you need it.

Without a helper

You keep repeating the same little dance instructions in every scene.

turn 15 degrees

move 10 steps

play sound “pop”

repeat it again...



With your own block

Make one instruction called **do a dance**. Put all the small steps inside it.

define do a dance

do a dance

do a dance

do a dance

Your project is easier to read, change, and share with a friend.

Open a new creative pack.

Extensions add extra blocks when your project needs a special power.



Click the blue Extensions button in Scratch to explore.



Music

Play notes and build rhythms.



Pen

Draw patterns as a sprite moves.



Text to Speech

Let a sprite speak typed words.



Translate

Try words in other languages.



Video Sensing

Use movement in front of a camera.



Device tools

Connect selected hardware in supported set-ups.

Pick the super block!

Point to an answer, then explain your reason. Ready? Go!

1. A character should walk across the stage. Which family helps?

- A. Motion
- B. Sound
- C. List

3. A game must remember a score that changes. Which family helps?

- A. Variables
- B. Pen
- C. Sound

2. The project must begin when the green flag is clicked. Which family helps?

- A. Looks
- B. Events
- C. Operators

4. A dance should happen four times. Which family helps?

- A. Sensing
- B. Control
- C. Translate



Teacher key

- 1 → Motion
- 2 → Events
- 3 → Variables
- 4 → Control

Build a Toolbox Story.

Your mission: Create a short interactive scene where a character moves, reacts to the player, and keeps track of one important thing.

Events

Start with a green flag or key.

Motion

Move your character.

Looks or Sound

Make the scene expressive.

Sensing

Notice an input or touch.

Variables

Keep a score, lives, or timer.

Deliverables + success

Choose a background and character. Add a start event, at least one movement, one reaction, and one changing variable. Your project should run from the green flag without needing a page reload.

Extension challenge

Add a repeat loop and an operator. For example, make a prize appear in a random place until the player reaches 10 points. Share the surprise you created with a partner.